EE326 Intro. to Information Theory and Coding

Midterm, Spring 2018

Date: Monday, Apr. 16, 2018 **Time:** 9am–12pm

Problem 1 (5 points)

Below you have five True/False questions. If you write a correct answer, you get 1 point for each question, but if you write an incorrect answer, you get -1 point for each question. If you don't write any answer, you get 0 point. (*You don't need to write down your reasonings for T/F questions.*)

a) For any $(X, Y) \sim P_{XY}$, it is ALWAYS true that for any $x \in \mathcal{X}$

$$H(Y) \geq H(Y|X = x).$$

True or False?

b) If $X \to Y \to Z$ and $X \to Z \to Y$, it is ALWAYS true that

$$I(X;Y) = I(X;Z).$$

True or False?

c) For any $(X, Y) \sim P_{XY}$, it is ALWAYS true that

$$I(X;Y) = D([P_X \cdot P_Y] \| P_{XY}).$$

True or False?

d) For any $(X, Y, Z) \sim P_{XYZ}$, it is ALWAYS true that

$$I(Z;Y) - I(Z;Y|X) = I(X;Z) - I(X;Z|Y).$$

True or False?

e) For any $(X, Y, Z) \sim P_{XYZ}$, it is ALWAYS true that

$$D(P_{XYZ} \| [P_X \cdot P_Y \cdot P_Z]) = H(X) + H(Y) + H(Z) - H(XYZ).$$

True or False?

Problem 2 (15 points)

In class, we defined the strong typical set, with respect to a particular distribution. In this problem, we generalize this to joint distributions. Let P_{XY} be a given joint distribution of X and Y, defined over alphabets $\mathcal{X}$ and $\mathcal{Y}$, respectively. We write P_X and P_Y as the corresponding marginal distributions of X and Y, respectively. For convenience, we write $W = P_{Y|X}$ as the conditional distribution. (*Write down your reasonings (or explicit calculations) for each question to get the full score.*)

a) (2 points) A pair of sequences (x^n, y^n) is said to have joint type of P_{XY} if its joint empirical distribution is P_{XY}, i.e.,

$$\begin{aligned}\hat{P}_{(x^n,y^n)}(a,b) &\triangleq \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}((x_i, y_i) = (a,b)) = \frac{1}{n}N((a,b)|(x^n,y^n)) \\ &= P_{XY}(a,b), \quad \forall a \in \mathcal{X}, \forall b \in Y.\end{aligned}$$

We write the type class, i.e., the collection of all such pairs of sequences as $T_{P_{XY}}$, i.e.,

$$T_{P_{XY}} = \{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n : \hat{P}_{(x^n,y^n)} = P_{XY}\}.$$

Compute the size $|T_{P_{XY}}|$ with exponential approximation (in $\doteq$ sense.)

b) (3 points) Now for each pair $(x^n, y^n) \in T_{P_{XY}}$, is it true that $x^n \in T_{P_X}$? Prove or disprove it.

c) (5 points) For a *given* $x^n \in T_{P_X}$, we define the conditional type

$$T_W(x^n) \triangleq \{y^n \in \mathcal{Y}^n : (x^n, y^n) \in T_{P_{XY}}\}.$$

We say such y^n sequences have the conditional type of W, given x^n. Compute the size of $T_W(x^n)$.
(Hint: You may use the definitions of types, i.e.,
1) For any $x^n \in T_{P_X}$, we have $N(a|x^n) = nP_X(a)$.
2) For any $(x^n, y^n) \in T_{P_{XY}}$, we have $N((a,b)|(x^n,y^n)) = nP_{XY}(a,b) = nP_X(a)W(b|a)$.)

d) (5 points) Now suppose we *independently* generate X^n sequence and Y^n sequence, according to the marginals P_X and P_Y, respectively. Compute the probability, with exponential approximation, that the resulting pair of sequences x^n and y^n, happens to satisfy that

$$x^n \in T_{P_X} \text{ and } y^n \in T_W(x^n).$$

(Hint: In class, we learned that $Q^n(T_P) \doteq e^{-nD(P\|Q)}$.)

Problem 3 (10 points)

Consider a pair of binary memoryless sources, (X_i, Y_i), $i = 1, 2, \ldots, n$, where $X_i \in \{0, 1\}$ and $Y_i \in \{0, 1\}$. Here, memoryless means that (X_i, Y_i) and (X_j, Y_j) are independent if $i \neq j$. Suppose (X_i, Y_i) has joint distribution P_{XY} for each i, and write the marginal distributions of X_i and Y_i as P_X and P_Y, respectively. Consider the source coding problem shown in the figure below. The encoder observes the source sequence X_i, $i = 1, 2, \ldots$ and tries to compress it into a bit stream of rate R bits/symbol, so that lossless reconstruction of the X sequence is possible at the decoder. At each time i, both the encoder and the decoder also observes Y_i, which serves as a "side information." The goal of this problem is to study how one should utilize this side information to reduce the code rate R.

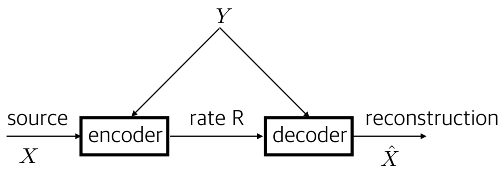

Figure 1: Source coding with side information

In this problem, you can assume that you have the optimal source code for any fixed distribution P, i.e., you can compress the source with rate $H(P)$ bits/symbol and reconstruct the source sequence from the codeword. By using this optimal source code as a building block, we want to design a coding scheme (encoder and decoder) that can compress the X sequence into $H(X|Y)$ bits/symbol when $n \to \infty$.

a) (2 points) Let the encoder first examines $y^n = (y_1, \ldots, y_n)$ sequence and constructs a sub-sequence of $x^n = (x_1, \ldots, x_n)$ by picking x_i's only at i's where $y_i = 0$. Let the corresponding sub-sequence be $x^{n(0)}$ where $n(0)$ is the number of 0's in y^n. ($x^{n(0)}$ is the length-$n(0)$ sub-sequence.) In a similar manner we also construct another sub-sequence $x^{n(1)}$ by picking x_i's only at i's where $y_i = 1$. Here $n(1)$ is the number of 1's in y^n. Note that $n(0) + n(1) = n$.

   Now we encode $x^{n(0)}$ and $x^{n(1)}$ separately, by using the optimal encoders for $P_{X|Y}(\cdot|0)$ and for $P_{X|Y}(\cdot|1)$, respectively, and just concatenate (link together) the codewords for $x^{n(0)}$ and $x^{n(1)}$ and let the decoder know this concatenated codeword. What is the total number of bits used to encode the length-n sequence x^n by this encoding scheme?

b) (3 points) Show that the average rate (bits/symbol) of the encoding scheme in a) goes to $H(X|Y)$ as $n \to \infty$.

c) (5 points) We just described the encoding scheme. Propose the corresponding decoding scheme that guarantees the reconstruction of x^n given the observed y^n sequence and the concatenated codeword for $x^{n(0)}$ and $x^{n(1)}$.

d) (Bonus: 5 points) Now assume that Y_i's are not given at time i but after a fixed delay D, i.e., the encoder and decoder observes Y_i at time $i + D$. For this new situation, design a coding scheme that can still achieve the rate $R = H(X|Y)$ as $n \to \infty$. (Now the encoder should encode $X_1, \ldots, X_n$ possibly by using $Y_{1-D}, \ldots, Y_{n-D}$, and the decoder should decode the binary codewords in $\{0, 1\}^{nR}$ back to $X_1, \ldots, X_n$ again by using $Y_{1-D}, \ldots, Y_{n-D}$.)